DSA EXPERIMENT 4

Database Management using arrays of structures

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Class:-SY1 Batch:-B1 PRN:B25ET1046 Div:-1

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input-

```
#include <stdio.h>
struct student
{
   int roll;
   char name[80];
   float SGPA;
};


void create(struct student arr[], int n);
void display(struct student arr[], int n);
void search(struct student arr[], int n);


void create(struct student arr[], int n)
{
   printf("Enter student details:\n");


   for (int i = 0; i < n; i++)
   {
      printf("Enter roll no, name and SGPA: ");
      scanf("%d %s %f",
            &arr[i].roll,
            arr[i].name,
            &arr[i].SGPA);
   }
}


void display(struct student arr[], int n)
{
   printf("\nStudent Details:\n");


   for (int i = 0; i < n; i++)
   {
      printf("Roll No: %d\n", arr[i].roll);
      printf("Name: %s\n", arr[i].name);
      printf("SGPA: %.2f\n", arr[i].SGPA);
```

```
    }
}

void search(struct student arr[], int n)
{
    int roll, i;

    printf("\nEnter roll number to search: ");
    scanf("%d", &roll);

    for (i = 0; i < n; i++)
    {
        if (arr[i].roll == roll)
        {
            printf("\nStudent found!\n");
            printf("Roll No: %d\n", arr[i].roll);
            printf("Name: %s\n", arr[i].name);
            printf("SGPA: %.2f\n", arr[i].SGPA);
            return;
        }
    }

    printf("\nStudent not found.\n");
}

int main()
{
    struct student s[100];
    int n;

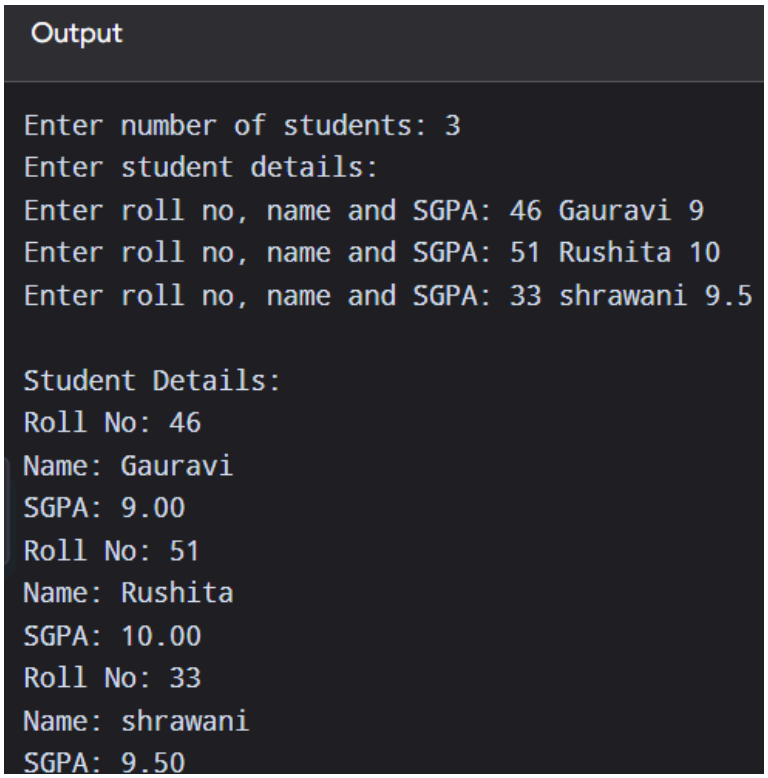
    printf("Enter number of students: ");
    scanf("%d", &n);

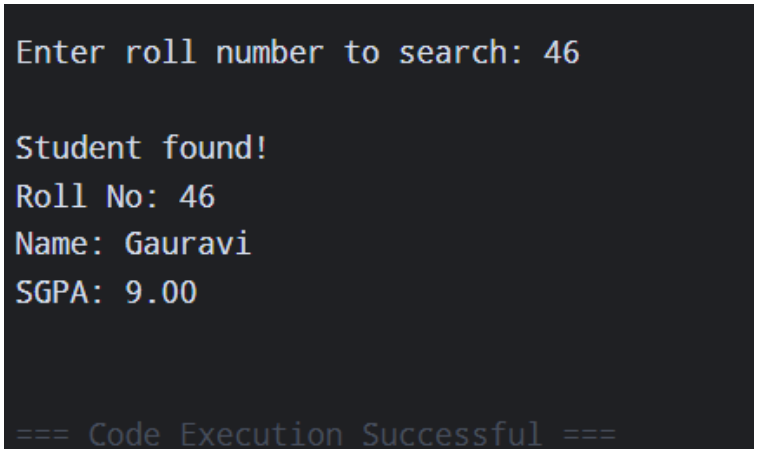
    create(s, n);
    display(s, n);
    search(s, n);

    return 0;
```

}

Output-


Output
Enter number of students: 3
Enter student details:
Enter roll no, name and SGPA: 46 Gauravi 9
Enter roll no, name and SGPA: 51 Rushita 10
Enter roll no, name and SGPA: 33 shrawani 9.5
Student Details:
Roll No: 46
Name: Gauravi
SGPA: 9.00
Roll No: 51
Name: Rushita
SGPA: 10.00
Roll No: 33
Name: shrawani
SGPA: 9.50


Enter roll number to search: 46
Student found!
Roll No: 46
Name: Gauravi
SGPA: 9.00
=== Code Execution Successful ===